Cointegration-Based Pairs Trading in Indian Bank Stocks

A Walk-Forward Evaluation

Arjun Prashanth*

October 2026

Abstract. I evaluate a cointegration-based pairs-trading strategy on 11 banks listed on India's National Stock Exchange. Every six months from January 2017 to June 2026, all 55 pairs are tested with the Engle-Granger procedure on the preceding twelve months, and up to five cointegrated pairs are traded over the next six months on a fixed z-score rule. Every parameter is estimated before trading begins, and costs of 20 basis points are charged on each leg. Over 2,347 out-of-sample days and 242 round trips, the strategy returns +1.27% a year before costs and −1.29% after, and the gross return is not distinguishable from zero (Newey-West $t = 0.67$). It is market-neutral, with a beta of −0.026 to the Nifty 50. Its weakness is persistence: 54% of trades are stopped out, more than converge, and the formation-period p-value does not predict which. The break-even cost is 9.8 basis points per leg.

Keywords: pairs trading; statistical arbitrage; cointegration; walk-forward backtest; transaction costs; market neutrality; Indian banks
JEL classification: C32, G11, G14, G17

1. Introduction

Pairs trading takes offsetting positions in two securities whose prices have historically moved together. When their relationship diverges, the strategy buys the relatively cheap security and sells the relatively expensive one, and it closes the position when the relationship reverts. Because the two legs offset, the position is designed to be insensitive to the direction of the market. Gatev et al. (2006) report average annualised excess returns of up to 11% for a version of the rule that matches US stocks by the distance between their normalised prices from 1962 to 2002. Do and Faff (2010) find that the profitability of the same rule declined substantially in later years, and Krauss (2017) reviews the main variants that have since been proposed.

The variant studied here selects pairs by cointegration (Engle and Granger, 1987), a stricter condition than correlation. Two prices can have highly correlated daily returns and still drift apart without limit if one has a slightly higher trend. If instead some linear combination $y_t - \beta x_t$ is stationary, deviations from its mean are expected to decay, which is the property a convergence trade depends on (Vidyamurthy, 2004). Avellaneda and Lee (2010) apply the same idea of trading mean-reverting residuals at the scale of the US equity market.

This paper asks whether that property survives out of sample in a setting where it is economically plausible: if pairs of Indian bank stocks are selected by a cointegration test on twelve months of prices and traded on a fixed rule over the following six months, does the strategy earn a positive return after costs, and is that return independent of the market? Banks share a regulator, an interest-rate cycle and a credit cycle, so a stable long-run relationship between their prices has an economic basis. Every parameter is estimated on data

*Email: arjunprashanth6129@gmail.com. Code, tests, results and the script that generates every number, table and figure in this paper are at https://github.com/arjunprashanth6129/pairs-trading-backtester. An interactive summary of the results is at https://arjun-quant-projects.vercel.app/pairs-trading.

that precede the period in which it is used, and the code that enforces this is covered by tests that fail if the separation is broken.

The strategy works as a hedge and fails as a source of return. Its beta to the Nifty 50 is close to zero, but its return before costs, +1.27% a year, is not statistically different from zero and is below the risk-free rate. A decomposition of the 242 trades explains why. Trades that converged did so, on average, at the speed the formation period predicted, but 130 of the 242 were stopped out after the spread widened further, and the losses on those trades cancelled the gains on the rest. The formation p-value carries no information about which pairs will hold. The largest single loss came from four simultaneous positions that were in effect one bet on a single bank.

The paper contributes a careful out-of-sample test of the cointegration approach in an emerging-market sector, a decomposition by exit type that identifies the mechanism behind the null result, and a record of an implementation error found during validation and how it changed the results (Appendix A). Section 2 describes the data and Section 3 the method. Section 4 presents the results, Section 5 discusses them and Section 6 sets out their limitations. Section 7 concludes.

2. Data

The universe is 14 banks listed on the National Stock Exchange. Daily closing prices adjusted for splits and dividends were downloaded from Yahoo Finance for 1 January 2016 to 31 July 2026. Three banks, RBL Bank (listed August 2016), AU Small Finance Bank (July 2017) and Bandhan Bank (March 2018), have price histories covering less than 95% of the sample and are excluded. The remaining 11 are HDFC Bank, ICICI Bank, Axis Bank, Kotak Mahindra Bank, State Bank of India, IndusInd Bank, Federal Bank, Bank of Baroda, Punjab National Bank, Canara Bank and IDFC First Bank, giving $\binom{11}{2} = 55$ candidate pairs. Only dates on which all 11 traded are retained. The out-of-sample period runs from 2 January 2017 to 30 June 2026, 2,347 trading days. The Nifty 50 index is used only to measure market exposure, and the risk-free rate is 6% a year.

3. Method

3.1 Formation and trading periods

The sample is divided into 19 windows, each a 12-month formation period followed by a 6-month trading period (Figure 1). Windows advance by six months, so the trading periods are contiguous, do not overlap, and together cover January 2017 to June 2026. Pairs are selected and every parameter is estimated in the formation period, and they are then held fixed for the whole trading period.

3.2 Pair selection

In each formation period, every candidate pair is tested with the two-step procedure of Engle and Granger (1987). The first step estimates the hedge ratio by ordinary least squares,

$$y_t = \alpha + \beta x_t + e_t, \tag{1}$$

and the second tests the residual e_t for a unit root with an augmented Dickey-Fuller test (Dickey and Fuller, 1979), with the lag length chosen by the Akaike information criterion and p-values from the response surfaces of MacKinnon (2010), as implemented in statsmodels (Seabold and Perktold, 2010). The null hypothesis is no cointegration. The dependent leg y is the stock with the higher average price, so that the orientation of each pair is fixed by rule rather than chosen. Pairs with $p \leq 0.05$ are retained, at most 5 per window and lowest p-value first.

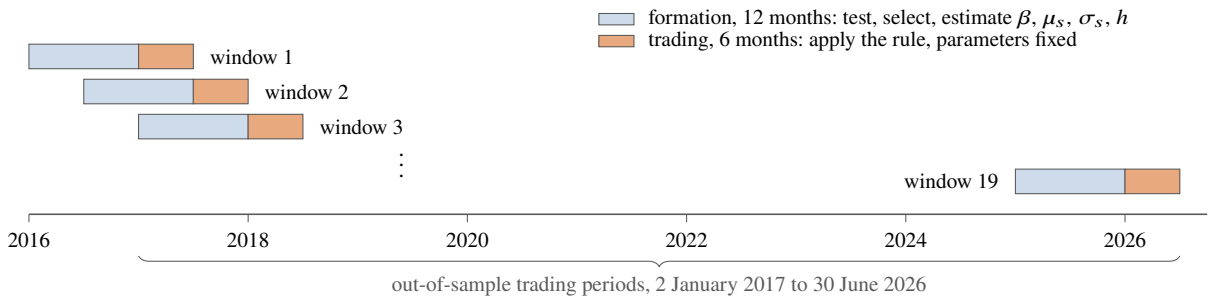


Figure 1. Formation and trading periods, drawn to scale. Each window's twelve-month formation period (blue) selects the pairs and fixes their parameters; the following six-month trading period (orange) applies them without re-estimation. Windows advance by six months, so trading periods tile the out-of-sample period. Only the first three and the last of the 19 windows are shown.

For each retained pair the formation period also fixes the spread mean μ_s, the spread standard deviation σ_s and the half-life of mean reversion. The half-life comes from a discrete approximation to an Ornstein-Uhlenbeck process,

$$\Delta s_t = a + \kappa\, s_{t-1} + u_t, \qquad h = -\frac{\ln 2}{\kappa}, \tag{2}$$

estimated on the formation spread. Across the sample, 77 pair-windows were selected from 41 distinct pairs of banks, with a median p-value of 0.015 and a median half-life of 6.8 trading days.

3.3 Trading rule

In the trading period the spread is $s_t = y_t - \beta x_t$ and its standardised value is

$$z_t = \frac{s_t - \mu_s}{\sigma_s}, \tag{3}$$

with β, μ_s and σ_s taken from the formation period throughout. Figure 2 sets out the rule. When z_t rises above +2 the strategy sells the spread (sells y and buys β units of x), and when it falls below -2 it buys the spread. A position is closed when $|z_t|$ returns within 0.5 (converged), when $|z_t|$ reaches 4 (stopped out), after 60 trading days (timed out) or at the end of the trading period. No position is opened while $|z_t| \geq 4$, and at most one position per pair is open at a time. Signals are computed from the close of day t and executed at the close of day $t + 1$. The thresholds are conventional values fixed before the backtest was run; they were not chosen by performance, which would introduce look-ahead at the level of the parameters.

3.4 Position sizing and costs

Each position has a gross notional of INR 100,000 (one lakh), the sum of the absolute values of the two legs. Capital is divided equally among the pairs selected for a window, and a pair without an open position contributes a zero return for the day. Costs are 20 basis points per leg on traded notional, made up of about 5 basis points of brokerage and taxes, 10 of half the bid-ask spread and 5 of slippage. Both legs are charged at entry and again at exit, so a round trip costs 40 basis points of gross notional, about INR 400.

3.5 Controls against look-ahead

A backtest that uses future information does not fail visibly; it reports better numbers. Three mechanisms keep trading-period data out of every decision. First, the hedge ratio, spread mean and spread standard deviation are computed by a single function that receives only formation data. Second, the standardisation in equation (3) takes μ_s and σ_s as arguments rather than computing them from its input, so a trading-period spread cannot

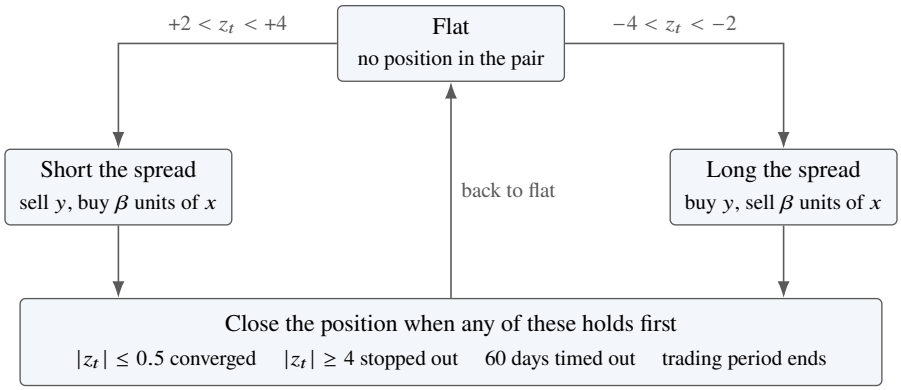


Figure 2. The trading rule. The z-score is computed from the close of day t using formation-period parameters, and every transition is executed at the close of day $t + 1$. Entry requires $|z_t|$ to lie between the entry and stop levels, so no position is opened while the spread is already beyond the stop. At most one position per pair is open at a time.

be standardised against its own statistics. Third, execution lags the signal by one day, because a fill at the close that produced the signal would assume knowledge of that close before it existed. Each mechanism is covered by unit tests, which perturb trading-period prices and check that the fitted parameters do not change, among other things; all 31 tests in the repository pass.

4. Results

4.1 Performance

Table 1 reports the strategy before and after costs alongside two long-only references measured over the same days. Over the out-of-sample period the strategy returns +1.27% a year before costs and −1.29% after, with annualised volatility of about 7.3%. The 242 round trips have a mean holding period of 15.9 trading days (median 11), and 52.1% are profitable before costs and 46.7% after. The mean trade earns INR 192 before costs, 19 basis points of its notional, and pays INR 403, or 40 basis points.

The gross return is not distinguishable from zero. The mean daily return before costs is 0.61 basis points, and with Newey-West standard errors (Newey and West, 1987) with ten lags its t-statistic is 0.67 ($p = 0.50$); after costs it is −0.46 ($p = 0.65$). At the trade level, the mean gross profit has a t-statistic of 0.70. The Sharpe ratio is negative even before costs, −0.65, because the gross return is below the 6% risk-free rate. A market-

Table 1. Out-of-sample performance

	Pairs strategy		Long-only references	
	Before costs	After costs	11 banks, equal weight	Nifty 50
Growth of 1	1.12	0.89	3.34	2.92
Annualised return (%)	+1.27	−1.29	+13.81	+12.18
Annualised volatility (%)	7.27	7.30	25.31	16.35
Sharpe ratio, $r_f = 6\%$	−0.65	−1.00	+0.31	+0.38
Maximum drawdown (%)	13.59	20.17	57.95	38.44

2 January 2017 to 30 June 2026, 2,347 trading days. The pairs strategy is reported before and after costs of 20 basis points per leg. The equal-weight reference holds the 11 traded banks; the Nifty 50 reference holds the index. Returns are compound annual returns.

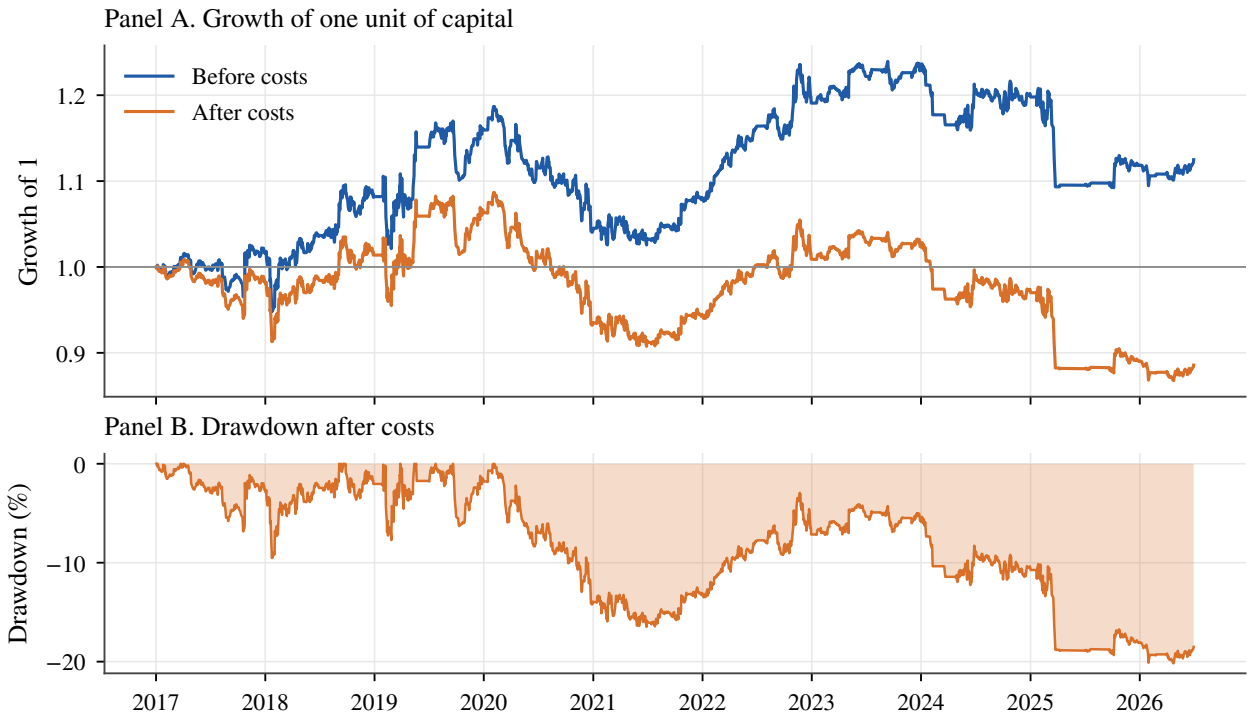


Figure 3. Out-of-sample growth of one unit of capital before and after costs (Panel A), and the drawdown of the after-cost series from its running peak (Panel B). The gap between the two series in Panel A is the cumulative cost of 242 round trips.

neutral portfolio would in practice earn interest on the collateral it posts, which this backtest does not credit (Section 6).

Figure 3 shows the path. The before-cost series rises to about 1.24 in September 2023 and then gives back about half of its gain, most of it in March 2025. That episode and an earlier decline in 2020 are examined in Section 4.6. The after-cost series has the same shape and ends 0.89 against 1.12.

4.2 An example pair

Figure 4 shows HDFC Bank against Kotak Mahindra Bank in the first trading window. The pair passed the formation test with $p = 0.017$ and a half-life of 6.4 days, and it is one of four pairs selected in four windows, more often than any other. The window contains every outcome the rule can produce. The first position opened just above +2, kept widening and was stopped out in late February. Three short positions, opened between late February and late April, each closed when the spread returned near zero. The last, opened in late May, was stopped out on the final day of the window as the spread rose above +4 again.

4.3 Decomposition by exit type

Table 2 divides the 242 trades by how they closed. The 71 trades that converged earned INR 320,605 before costs. The 130 that were stopped out lost INR 305,070. The other two exits are small. Stopped trades therefore outnumbered converged trades by 1.8 to one, and their losses almost exactly cancelled the gains (Figure 5, Panel A). Costs then turned a small gross profit into a loss.

The averages make the arithmetic explicit. A converged trade earned INR 4,516 on average and a stopped trade lost INR 2,347. At those payoffs, converged trades must make up at least 34.2% of converged and stopped trades combined for the strategy to break even before costs. The realised share is 35.3% (71 of the

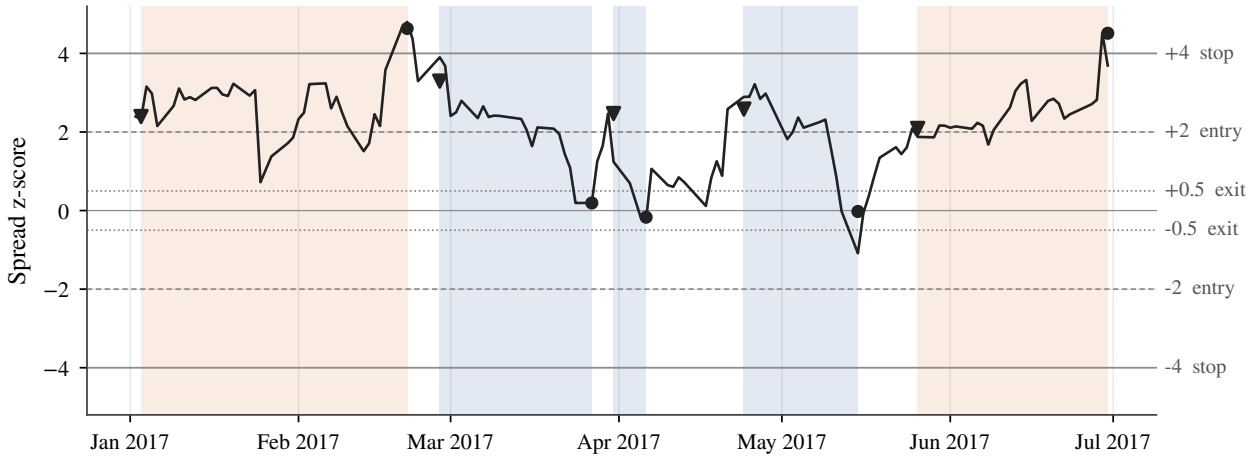


Figure 4. HDFC Bank against Kotak Mahindra Bank, January to June 2017. The line is the spread's z-score computed with formation-period parameters. Shaded spans are positions, blue for those closed at a profit after costs and orange for those closed at a loss; triangles mark entries and circles exits. Markers are plotted at execution, one day after the signal, so some entries fall inside the entry band.

Table 2. Trades by exit type

Exit	Trades	Share (%)	Gross P&L	Costs	Net P&L	Median days held
Converged ($\lvert z\rvert \le 0.5$)	71	29.3	320,605	28,477	292,128	14
Stopped out ($\lvert z\rvert \ge 4$)	130	53.7	−305,070	52,401	−357,471	3.5
Timed out (60 days)	8	3.3	10,173	3,243	6,930	60
Trading period ended	33	13.6	20,660	13,365	7,295	25
Total	242	100.0	46,368	97,486	−51,118	11

Profit and loss in rupees, summed over all trades in each row, for positions with a gross notional of INR 100,000. Costs are 20 basis points per leg, charged on both legs at entry and exit.

201 converged or stopped trades), which is not significantly different from that threshold in a binomial test ($p = 0.77$). That is why the gross return is close to zero. A stop at $|z| = 4$ means the spread moved a further two formation standard deviations against the position, and 68 of the 130 stopped trades reached that level within four trading days of entry. The median stopped trade lasted 3.5 days, against 14 for a converged trade (Figure 5, Panel B). For more than half of all trades, a relationship that passed a cointegration test over the previous twelve months did not hold over the following weeks.

4.4 Speed of convergence

The formation half-life predicts how fast a deviation should decay. For the 71 converged trades, the median holding period is 14 trading days and the median formation half-life of the corresponding pairs is 6.4 days, a ratio of 1.98. That ratio is close to its theoretical value. The median converged trade entered at $|z| = 2.27$ and exited at 0.5, so the deviation had to shrink by a factor of 4.5, which takes $\log_2 4.5 \approx 2.2$ half-lives under an autoregressive process.

Figure 6, Panel A, makes the comparison trade by trade. It plots each converged trade's holding period against the period predicted by its formation half-life h and entry level, $h \log_2(|z_{\text{entry}}|/0.5)$. The median ratio of realised to predicted is 0.86, so the half-life gets the typical speed about right, but it says little about individual trades: realised holding periods run from 2 to 59 days, and their correlation with the prediction is

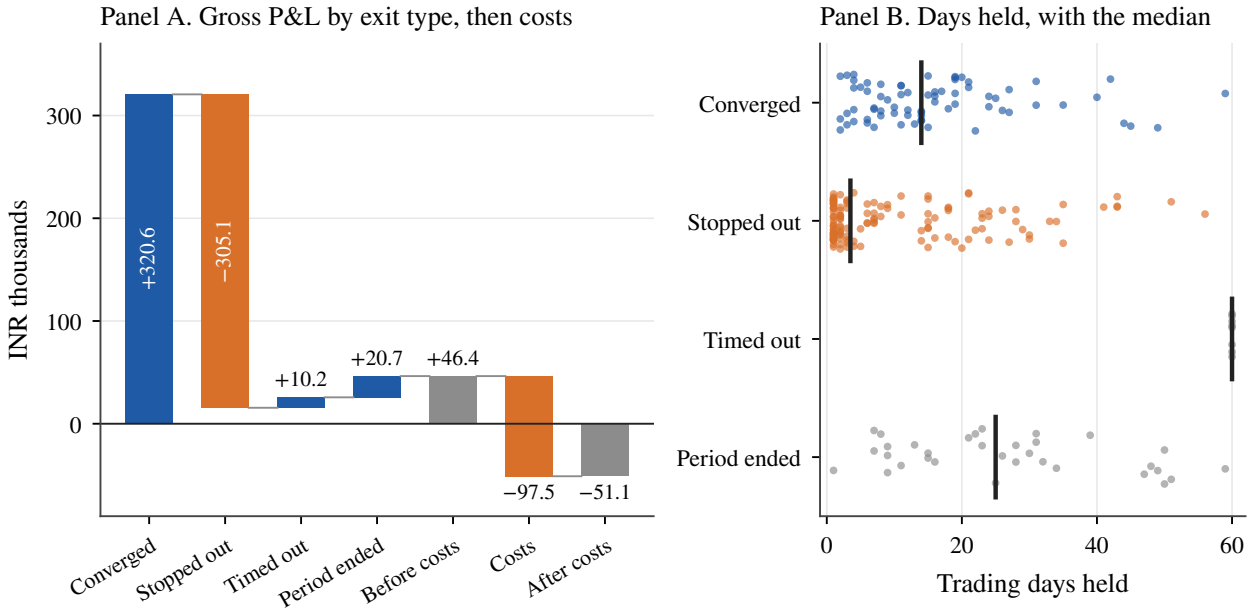


Figure 5. Trades by exit type. Panel A builds the gross profit and loss from the four exit types, in thousands of rupees, then deducts costs. Panel B shows how many trading days each trade was held, one point per trade, with the median of each exit type as a vertical bar. Trades that time out are all held for 60 days.

Table 3. Trade outcomes by formation-period p-value

Formation p-value	Trades	Stopped out (%)	Converged (%)	Mean gross P&L
0.00 to 0.01	93	59.1	26.9	85
0.01 to 0.03	89	46.1	33.7	302
0.03 to 0.05	60	56.7	26.7	192

Trades grouped by the Engle-Granger p-value of their pair in the formation period. Mean gross profit and loss per trade is in rupees.

0.12. On average, then, the estimate of speed held out of sample. The assumption that the spread would keep reverting did not hold for most trades.

4.5 Formation p-value and trade outcome

If the formation p-value measured how reliable a pair's relationship would be, a smaller p-value should come with fewer stop-outs. Table 3 and Figure 6, Panel B, show that it does not: the group with the smallest p-values has the highest stop-out rate. A χ^2 test of independence between the p-value group and stopping out gives $\chi^2(2) = 3.40$ with $p = 0.18$. Among pairs that pass the 5% threshold, the p-value carries no detectable information about outcomes in the following six months, so ranking pairs by it adds nothing beyond the threshold itself.

Repeated selection is no better a guide. A pair chosen in several windows has passed the test on several different years of prices, which might be expected to signal a durable relationship. Table 4 lists the seven pairs selected in three or more windows. Of these, four lost money before costs, and 51.9% of their trades were stopped out, against 53.7% for all trades.

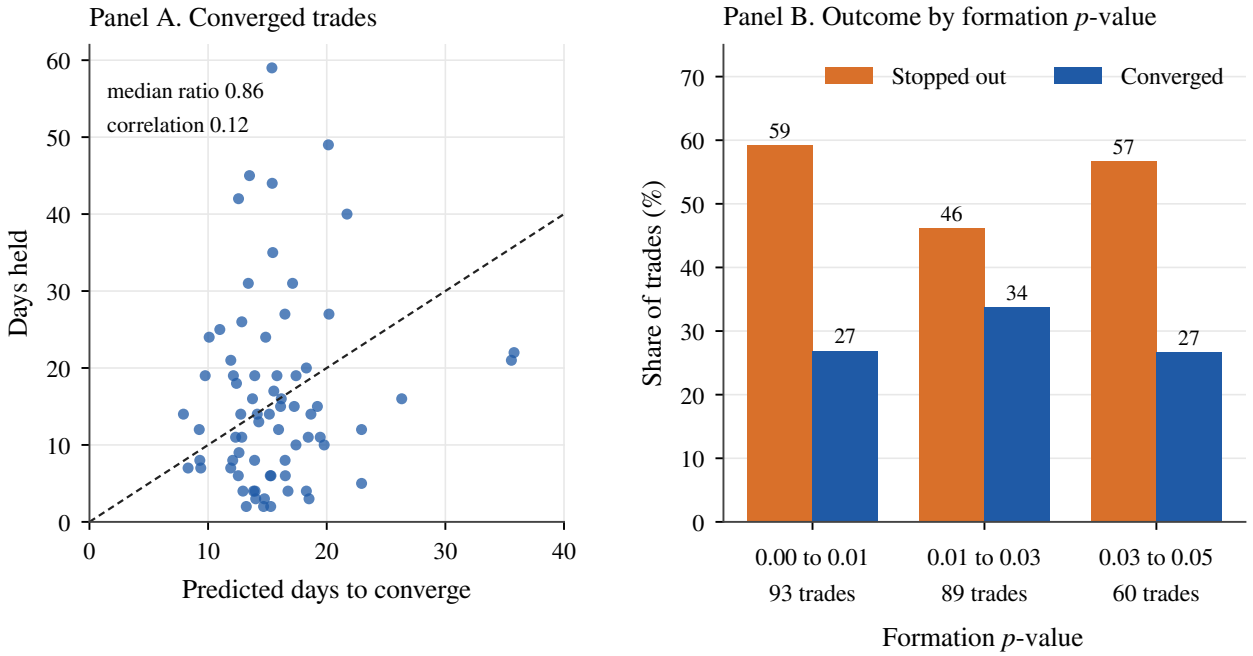


Figure 6. What the formation period predicts. Panel A plots each of the 71 converged trades' holding period against the period predicted by its formation half-life and entry z-score; the dashed line marks equality. Panel B shows the share of trades stopped out and converged in each group of formation *p*-values; the remaining trades timed out or were open at the end of the trading period.

Table 4. Pairs selected in three or more windows

		Formation, median		Trades			
Pair	Windows	*p*-value	Half-life	All	Converged	Stopped	Gross P&L
HDFC Bank / Kotak Mahindra Bank	4	0.018	6.7	18	8	6	23,297
Canara Bank / Punjab National Bank	4	0.013	9.9	9	3	3	4,861
Bank of Baroda / Federal Bank	4	0.002	6.2	13	4	8	−722
IDFC First Bank / IndusInd Bank	4	0.003	5.4	13	1	9	−22,589
HDFC Bank / IndusInd Bank	3	0.019	5.0	11	7	3	28,796
HDFC Bank / ICICI Bank	3	0.010	7.8	9	1	7	−3,400
ICICI Bank / State Bank of India	3	0.009	7.8	6	1	5	−4,605

A pair is identified by its two banks; which bank is the dependent leg can differ between windows. The *p*-value and half-life (in trading days) are medians over the windows in which the pair was selected. Gross profit and loss is summed over the pair's trades, in rupees, before costs.

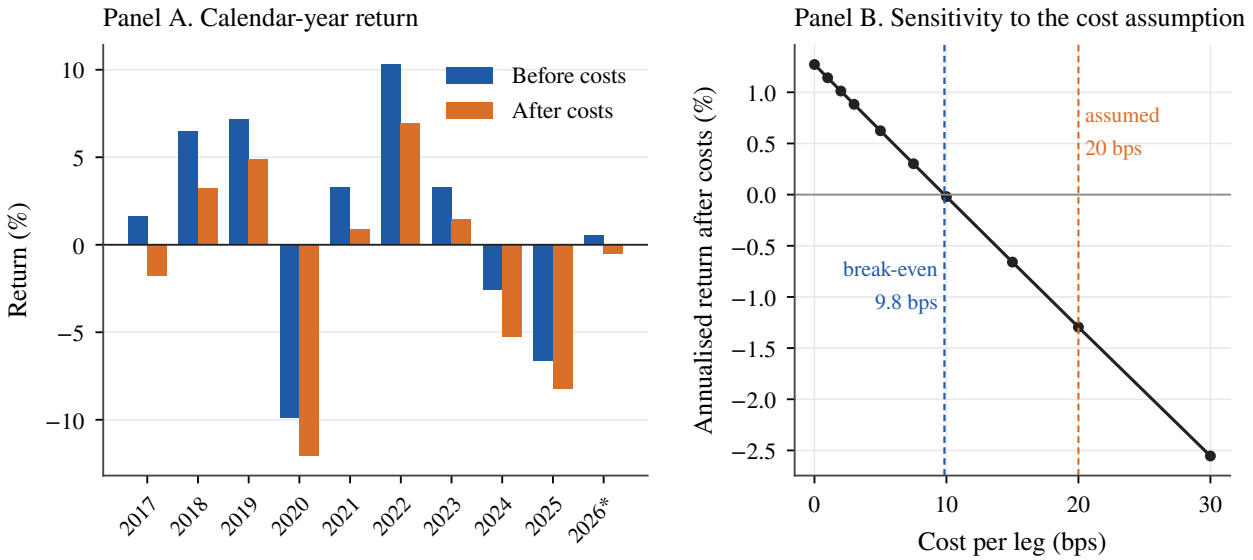


Figure 7. Panel A shows the return in each calendar year before and after costs; 2026 (marked *) is a partial year. Panel B shows the annualised after-cost return when the same trades are repriced at different costs per leg. Costs are linear in the rate, so every point is exact, and the value at the assumed 20 basis points equals the backtest's after-cost return.

4.6 Performance by period and concentration

Before costs the strategy was profitable in 6 of 9 full calendar years and in 12 of 19 trading windows (Figure 7, Panel A). The losses are concentrated in a few episodes. In 2020 the strategy lost −9.86% before costs. During the COVID-19 sell-off bank stocks fell by very different amounts, and two pairs account for most of the loss: Axis Bank against Federal Bank was stopped out six times for INR 20,810, and IndusInd Bank against IDFC First Bank lost INR 20,500 over seven trades.

The single largest loss occurred in the window that began trading on 1 January 2025. Four of the five pairs selected on the 2024 formation period had Kotak Mahindra Bank as a leg, and on 21 January 2025 all four were entered on the same day, each short Kotak. The three positions against public-sector banks, Bank of Baroda, Canara Bank and Punjab National Bank, were stopped out together on 24 March 2025 for about INR 12,500 each, and the position against Federal Bank on 20 March 2025 for INR 7,236. Together the four lost INR 44,660 before costs, about as much as the strategy's entire gross profit of INR 46,368 over nine and a half years. The four positions were, in effect, a single exposure to Kotak relative to other banks. The selection rule places no limit on how many selected pairs share a stock, and in 7 of the 19 windows one bank was a leg of three or more selected pairs.

4.7 Transaction costs

The signals do not depend on costs, so the same 242 trades can be repriced at any cost level (Figure 7, Panel B). The break-even follows from the trade statistics: the mean trade earned 19 basis points of notional before costs, and a round trip costs $2c$ basis points at c basis points per leg, so $c \approx 9.6$; on the compounded daily series the break-even is 9.8 basis points per leg. That is about half the assumed cost and within the range of institutional execution in large Indian banks, but below typical retail costs. Even at zero cost the strategy would return +1.27% a year, below the risk-free rate.

4.8 Market exposure

Regressing the strategy's daily after-cost return on the Nifty 50 gives a beta of −0.026 and a correlation of −0.057; the market explains 0.3% of the strategy's daily variance. A 1% move in the index is associated with a move of about 0.03% in the strategy. The long-short construction removed market exposure as intended, and what remains is a bet on relative price movements between banks.

5. Discussion

The central result concerns persistence. The formation test selects pairs whose spreads revert at a measurable speed, and when those spreads converged in the trading period they did so, on average, at almost exactly that speed. But more than half of the trades were stopped out after the spread widened further, and neither the formation p-value nor repeated selection predicted which. Passing a cointegration test over one year is evidence that a relationship held over that year; in this sample it was weak evidence that the relationship would hold over the next six months.

The gross return is consistent with zero. Lower costs would raise the after-cost return towards the before-cost return, but not above the risk-free rate. This fits the decline in the profitability of pairs trading reported by Do and Faff (2010), and it suggests that, among the most liquid Indian banks, relationships stable enough to trade are either rare or already traded.

Concentration produced the largest loss. A limit on the number of selected pairs that may share a stock would have turned four correlated positions into two, and is the simplest change likely to reduce the strategy's worst outcomes. The market neutrality of the construction, by contrast, worked exactly as intended.

6. Limitations

- *One sector and one market.* All eleven stocks are Indian banks, and the results need not extend to other sectors or markets.
- *Financing.* No interest is credited on collateral, which would raise both return series and make the Sharpe ratio less negative without changing the trade-level results. Holding a short equity position in India for more than a day requires stock lending or single-stock futures, whose basis and roll costs are not modelled.
- *Costs and execution.* The cost of 20 basis points per leg does not vary with volatility or trade size and is probably too low on the days when spreads widen sharply, which are the days stops are triggered. All trades are executed at closing prices.
- *Parameters.* The thresholds, the 60-day limit and the twelve-month formation period are conventional values. Testing alternatives properly would require a further holdout period.
- *Survivorship.* The universe consists of banks listed in 2026. Banks merged during the sample, such as those absorbed in the 2020 public-sector mergers, were never candidates.
- *Sample sizes.* Several comparisons rest on 60 to 130 trades per group and should be read with that in mind.

7. Conclusion

A cointegration-based pairs strategy on Indian bank stocks, with every parameter fixed before it is used, is market-neutral but does not earn a return distinguishable from zero before costs, and it loses money after them. The reason is visible in the trades: relationships that passed a cointegration test over twelve months broke down within weeks more often than they converged, and the strength of the test statistic did not identify

which would hold. The most valuable extensions address that directly: re-testing cointegration on a rolling window during the trading period and closing a pair when the test fails, limiting how many selected pairs may share a stock, estimating the hedge ratio with a Kalman filter, and using the procedure of Johansen (1988) for baskets of more than two stocks.

Use of AI tools

The research question, the design of the study, the trading rules and tests, and the interpretation of the results are my own. I used Claude Code, Anthropic's AI coding assistant, in two supporting roles. The first was as a tutor. When the work reached an idea I had not used before, such as why a convergence trade needs cointegration rather than correlation, I asked it to explain the idea and then checked the explanation against the data before relying on it. The second was implementation: at my direction it wrote parts of the code, including the plotting code and the script that generates this paper's numbers, tables and figures, and it helped me draft and edit the text. Working this way also taught me how to use an AI assistant well, by specifying a task precisely, asking for the reasoning behind a suggestion and not only the code, and verifying every output. I reviewed everything it produced and take full responsibility for the content.

Data and code availability

Code, tests and results are available in the project repository, https://github.com/arjunprashanth6129/pairs-trading-backtester. Running the pipeline regenerates every result from prices it downloads and caches on the first run, and `python paper/make_assets.py` regenerates every number, table and figure in this paper from the committed results.

References

Avellaneda, M. and J.-H. Lee (2010). Statistical arbitrage in the US equities market. *Quantitative Finance 10*(7), 761–782.

Dickey, D. A. and W. A. Fuller (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association 74*(366), 427–431.

Do, B. and R. Faff (2010). Does simple pairs trading still work? *Financial Analysts Journal 66*(4), 83–95.

Engle, R. F. and C. W. J. Granger (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica 55*(2), 251–276.

Gatev, E., W. N. Goetzmann, and K. G. Rouwenhorst (2006). Pairs trading: Performance of a relative-value arbitrage rule. *Review of Financial Studies 19*(3), 797–827.

Johansen, S. (1988). Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control 12*(2–3), 231–254.

Krauss, C. (2017). Statistical arbitrage pairs trading strategies: Review and outlook. *Journal of Economic Surveys 31*(2), 513–545.

MacKinnon, J. G. (2010). Critical values for cointegration tests. Working Paper 1227, Queen's University, Department of Economics.

Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica 55*(3), 703–708.

Seabold, S. and J. Perktold (2010). Statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference*, pp. 92–96.

Vidyamurthy, G. (2004). *Pairs Trading: Quantitative Methods and Analysis*. Hoboken, NJ: Wiley.

A. Correction of an entry-rule error

The first complete run of the backtest reported 1,008 trades and an after-cost return of −9.82% a year. A breakdown by exit type showed 892 stop-outs with a median holding period of one day. The entry rule had opened a position whenever $|z_t| \geq 2$ without checking whether $|z_t|$ also exceeded the stop level. When a spread moved beyond the stop, the strategy entered, was stopped out the next day and entered again, for as long as the spread stayed beyond the stop, paying four legs of cost on each cycle.

Table 5. The backtest before and after the correction

	Before correction	After correction
Round trips	1,008	242
Entered with $\|z\| \geq 4$ already	816	0
Stopped out	892	130
Median holding period (days)	1	11
Annualised return before costs (%)	+1.04	+1.27
Annualised return after costs (%)	−9.82	−1.29
Maximum drawdown after costs (%)	62.16	20.17
Break-even cost per leg (bps)	1.9	9.8

The "before" column is computed from the results committed at revision `00591ed`, the last revision before the correction; the "after" column is the current backtest.

The correction adds one condition, no entry while $|z_t| \geq 4$, and two tests cover it. A position may be opened once the spread returns inside the stop level. The error changed the after-cost return by 8.5 percentage points a year and the before-cost return by only 0.2. Because its effect appeared almost entirely as cost, the first run looked like a strategy with a gross edge that could not cover its costs. The decomposition by exit type showed that this reading was wrong, and it is the reason the decomposition appears in the main results.